

François ROZET

PhD · Machine Learning · Scientist · Engineer

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 FrancoisRozet

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Education

PhD in Deep Learning

 Sep 2021 – Jan 2026  University of Liège  Belgium

The goal of my PhD, completed under the supervision of Prof. Gilles LOUPPE, was to develop and apply deep learning methods to Bayesian inference problems in large-scale dynamical systems (oceans, atmospheres, molecules, ...). The contributions presented in my thesis lie at the intersection of many subjects, most notably generative modeling, inverse problems and physics emulation.

MSc in Data Science and Engineering

 Sep 2019 – Jul 2021  University of Liège  Belgium

My Master's degree program was built around data technologies, artificial intelligence and their mathematical and computational foundations. I graduated *summa cum laude* with the congratulations of the examination committee (top 1%) and received two awards for my Master's thesis, titled "Arbitrary Marginal Neural Ratio Estimation for Likelihood-free Inference".

Work

Research Internship

 Sep 2024 – Jan 2025  Flatiron Institute  New York, USA

The goal of this internship was to improve and accelerate physics simulations using deep learning methods, including latent representation techniques and generative models. The associated publication is currently under peer review.

Publications

- [1] **Rozet**. "Generative Modeling in Large-scale Dynamical Systems", 2026. URL: <https://orbi.uliege.be/handle/2268/337979>
- [2] Andry, **Rozet**, Lewin, *et al.* "Appa: Bending Weather Dynamics with Latent Diffusion Models for Global Data Assimilation", 2025. URL: <http://arxiv.org/abs/2504.18720>
- [3] **Rozet**, Ohana, McCabe, *et al.* "Lost in Latent Space: An Empirical Study of Latent Diffusion Models for Physics Emulation", 2025. URL: <http://arxiv.org/abs/2507.02608>
- [4] Ohana, McCabe, Meyer, *et al.* "The Well: a Large-Scale Collection of Diverse Physics Simulations for Machine Learning" in *Advances in Neural Information Processing Systems*, 2024. URL: <https://openreview.net/forum?id=00Sx577BT3>
- [5] **Rozet**, Andry, Lanusse, and Louppe. "Learning Diffusion Priors from Observations by Expectation Maximization" in *Advances in Neural Information Processing Systems*, 2024. URL: <https://openreview.net/forum?id=7v88Fh6iSM>
- [6] **Rozet** and Louppe. "Score-based Data Assimilation" in *Advances in Neural Information Processing Systems*, 2023. URL: <https://openreview.net/forum?id=VUvLSnMZdX>
- [7] **Rozet** and Louppe. "Score-based Data Assimilation for a Two-Layer Quasi-Geostrophic Model" in *Machine Learning and the Physical Sciences Workshop (NeurIPS)*, 2023. URL: <http://arxiv.org/abs/2310.01853>
- [8] Vasist, **Rozet**, Absil, *et al.* "Neural posterior estimation for exoplanetary atmospheric retrieval" in *Astronomy & Astrophysics*, 2023.
- [9] Delaunoy, Hermans, **Rozet**, *et al.* "Towards Reliable Simulation-Based Inference with Balanced Neural Ratio Estimation" in *Advances in Neural Information Processing Systems*, 2022. URL: <https://openreview.net/forum?id=o762mMj4XK>
- [10] Hermans, Delaunoy, **Rozet**, *et al.* "A Trust Crisis In Simulation-Based Inference? Your Posterior Approximations Can Be Unfaithful" in *Transactions on Machine Learning Research*, 2022. URL: <https://openreview.net/forum?id=LHAbHkt6Aq>
- [11] **Rozet** and Louppe. "Arbitrary Marginal Neural Ratio Estimation for Simulation-based Inference" in *Machine Learning and the Physical Sciences Workshop (NeurIPS)*, 2021. URL: <http://arxiv.org/abs/2110.00449>

Talks

- *Learning Diffusion Priors from Observations by Expectation Maximization* – Tübingen AI Center (Jul 2024)
- *Diffusion Models meet Data Assimilation* – ENGIE (Jul 2024)
- *Score-based Generative Models meet Data Assimilation* – ESA-ECMWF Workshop (May 2024)

Community

Reviewing

Since 2022, I am a recurrent peer-reviewer for machine learning journals, conferences and workshops such as NeurIPS, ICML, ICLR, AISTATS and TMLR.

Honors & Awards

- Research fellow grant – F.R.S.-FNRS (2022)
- Best Master's thesis award – AIM (2021)
- Best Master's thesis award – AILg & NRB (2021)
- National Physics olympiad 3rd laureate – ABPPC (2016)

Software

Azula

 July 2024  probabilists/azula ★ 128

Azula is a Python package that implements diffusion models in PyTorch. Its goal is to unify the different formalisms and notations of the generative diffusion models literature into a single, convenient and hackable interface.

Inox

 Dec 2023  francois-rozet/inox ★ 34

Inox is a minimal JAX library for neural networks with an intuitive PyTorch-like syntax. Modules are represented as PyTrees, which enables complex architectures, easy manipulations, and functional transformations.

Zuko

 Oct 2022  probabilists/zuko ★ 450

Zuko is a Python package that implements normalizing flows in PyTorch. It relies heavily on PyTorch's built-in distributions and transformations, which makes the implementation concise, easy to understand and extend.

PIQA

 Oct 2020  francois-rozet/piqa ★ 434

PIQA is a collection of PyTorch metrics for image quality assessment in various image processing tasks such as generation, denoising, super-resolution, interpolation, etc. It focuses on the efficiency, conciseness and understandability of its (sub-)modules, such that anyone can easily reuse and/or adapt them to its needs

Languages

French native
English proficient (C1, TOEFL 106, PTE 90)
Dutch elementary (A1)

Skills

Python  (NumPy, PyTorch, JAX, ...), C/C++, Julia, HTML ,
JavaScript , Git , GitHub , Linux , Docker , Slurm

Teaching

Teaching assistant

 Sep 2021 – Present  University of Liège

- Introduction to artificial intelligence – Prof. Gilles LOUPPE (2021-2025)
- Deep learning – Prof. Gilles LOUPPE (2021-2025)
- Data structure and algorithms – Prof. Pierre GEURTS (2021-2022)
- Functional programming – Prof. Christophe DEBRUYNE (2021-2022)